Yufeng (Alex) Xia

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COMPUTER SCIENCE STUDENT · RESEARCH ASSISTANT

Education

- **University of Edinburgh** — Edinburgh, UK
  MSc by Research in Computer Science, supervised by Prof. Mahesh K. Marina — *Sep 2025 – Present*
- **University of Edinburgh** — Edinburgh, UK
  Bachelor of Science in Computer Science; Grade: First Class (Hons) — *Sep 2021 – May 2025*

Self Description

- I am a self-motivated and detail-oriented researcher with a strong passion for exploring innovative solutions to complex problems in compiler technologies, decentralized learning, and embedded systems optimization. I enjoy conducting independent research, collaborating with multidisciplinary teams, and bridging theoretical knowledge with practical applications. My goal is to contribute to advancing technology through rigorous analysis, experimentation, and creative problem-solving.

Skills

- **Programming Languages**: Advanced proficiency in C, C++, Python, Haskell, MIPS. Specialized in Java for network programming and troubleshooting.
- **Research Skills**: Demonstrated ability to conduct independent and collaborative research, from problem identification to solution implementation. Experienced in designing experiments, analyzing results, and integrating theoretical knowledge with practical applications. Skilled in effectively communicating research findings through presentations and publications.
- **Development Tools**: Proficient in using Git for version control, Maven for project management, and Docker for containerization, ensuring efficient development workflows.
- **Web Development**: Proficient in building scalable and maintainable web applications using modern frameworks and technologies. Experienced in RESTful API design, microservices architecture, and containerized deployments with Docker and Kubernetes.
- **Cloud-Native Technologies**: Hands on experience in cloud-native development, leveraging CI/CD pipelines, serverless computing, and infrastructure as code (IaC) for automated and efficient deployment of applications.
- **Compiler Technologies**: Familiar with LLVM, GCC, LLVM Pass development, and compiler optimizations, including IR-level transformations and platform-specific tuning. Skilled in platform-specific tuning for embedded architectures. Experienced in debugging and profiling compiler-generated code to identify bottlenecks and improve runtime efficiency.
- **Machine Learning & AI**: Experienced in decentralized and federated learning, with a focus on enhancing privacy and security in distributed systems. Skilled in using frameworks like PyTorch and TensorFlow for distributed model training and privacy-preserving applications. Actively engaged in virtual seminars and workshops led by leading academics.
- **Linux**: Experienced in the Linux environment with expertise in system administration, shell scripting, and environment configuration. Proficient in Linux kernel programming, including module development, debugging, and performance tuning for custom operating system functionality.

Experience

- **University of Edinburgh** — Edinburgh, UK
  Junior Research Assistant — *March 2024 - now*
  - **Compiler Optimization Research**: Junior Research Assistant at Institute for Computing Systems Architecture (ICSA), under supervision of Dr. Antonio Barbalace.
  - **Embedded Systems**: Conducting research on compiler optimization techniques to improve performance and enable efficient code generation for embedded systems.
  - **Cross-disciplinary Projects**: Collaborating on cross-disciplinary projects involving compilers, and hardware-software co-design.
  - **Publications**: Authoring technical reports, presenting research findings at internal meetings, and preparing manuscripts for academic publications in the domain of compilers and system architectures.

- **Senior Tech, Suzhou** Suzhou, China
  Back-end Development Intern and Software Tester *June 2023 - September 2023*
  - **API Development**: Designed and implemented clean, maintainable, and scalable APIs for backend services, ensuring seamless data exchange and request handling.
  - **Code Quality**: Participated in code reviews and debugging, working closely with developers to enhance system robustness and code quality.
  - **Platform Modernization**: Modernized Uniqlo's delivery platform by transitioning from JFinal to Spring Boot, significantly improving system performance, scalability, and maintainability.
  - **Android Testing**: Contributed to the Uniqlo Android App testing, ensuring high usability.
- **Carnegie Mellon University - Decentralized Machine Learning Research Group** Remote
  Research Participant *Jan 2024 - May 2024*
  - **Decentralized Learning**: Participated in cutting-edge research on decentralized machine learning, focusing on enhancing privacy and security in distributed systems.
  - **Academic Engagement**: Participated in virtual seminars and workshops led by leading academics in the field, contributing to discussions on the latest developments and challenges in decentralized learning.
  - **Practical Application**: Completed projects and case studies applying decentralized machine learning algorithms.

Projects

- **Cosense Auto-Quantization based on sensor's specification** Edinburgh, UK
  Junior Research Assistant *April 2024 - now*
  - **CoSense Compiler**: CoSense is a compiler that uses sensor information from datasheet to help with optimization. It is developed based on LLVM.
  - **LLVM Pass Development**: Currently developing and implementing an LLVM pass for the CoSense compiler, aiming to use sensor information from datasheets to optimize performance and power efficiency.
  - **Tech Stack**: LLVM, Compiler, Compiler Optimization, Performance Benchmarking
- **Uniqlo Express Delivery Platform** Senior Tech, Suzhou
  Software Engineer Intern *June 2023 - September 2023*
  - **Interface Development**: According to product requirements and development requirements, write easy maintenance and readable interface code for delivery platform system.
  - **Code Review**: Work regularly with the development team on code review to improve system code robustness.
  - **Platform Modernization**: Modernized and improved the efficiency of Uniqlo's express delivery platform by transitioning from JFinal to Spring Boot, enhancing performance and user experience.
  - **Business Logic**: Refined business logic and introduced visual aids like process and Gantt charts for clear visualization of project progress.
  - **Tech Stack**: Spring Boot, MySQL, Redis, Kafka, OpenResty, Documentation
- **Uniqlo Android App** Senior Tech, Suzhou
  Software Engineer Intern *June 2023 - September 2023*
  - **Testing**: Contributed to enhancing the primary mobile touchpoint for Uniqlo's customers, focusing on rigorous testing for performance, user experience, and reliability.
  - **Bug Resolution**: Proactively identified and resolved numerous bugs, significantly improving the app's usability and customer satisfaction.
  - **Skills**: Documentation, Communication, Software Testing, Agile Development
- **PizzaDronz - Pizza Delivery Management Software** School of Informatics, University of E
  Project Lead *September 2023 - Dece*
  - **Order Processing**: Developed the system's order processing and validation logic, integrating complex algorithms to ensure accuracy and efficiency in order handling.
  - **Pathfinding**: Fine-tuned the A* pathfinding algorithm for drone navigation, optimizing delivery routes and avoiding no-fly zones.
  - **CI/CD Pipeline**: Designed and implemented a robust CI/CD pipeline using GitHub Actions for streamlined development and high code quality.
  - **Tech Stack**: Java, Maven, RESTful services, Database, A* Pathfinding, CI/CD

Hobbies

- **Linux Enthusiasm**: Deeply passionate about kernel development and a committed Linux user. I have a strong preference for Vim and enjoy using Arch and Fedora distributions. My hobbies include customizing development environments and delving into the complexities of Linux systems, contributing to open-source projects, and enhancing my understanding of kernel architecture.
- **Table Tennis**: An avid table tennis player, actively participating in school club activities and contributing to team dynamics and skill development.
- **Cycling and Fitness**: Enthusiastic about staying active and healthy through cycling and fitness routines, enjoying the challenge and refreshment it brings to my daily life.

Languages

- Mandarin: Native Speaker — English: Fluent

Test Scores

- **GRE**: 336 / 340 (Jun 2024) — Verbal Reasoning 166/170, Quantitative Reasoning 170/170
- **IELTS**: 7.0

Research and Publications

- **Morphling: Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
  *Leyang Xue, **Yufeng Xia**, Eren Mendi, Ismaeel Bashir, Jiaxun Yang, Myungjin Lee, Mahesh Marina*
  - **Scalable Edge ML Emulator**: Developed Morphling, a measurement-driven emulator that evaluates distributed ML training on heterogeneous edge devices at scale, supporting up to 1800 emulated devices on a single GPU server with minimal latency and thermal error.
- **Weaver: Foundation Model Training over AI-RAN Compute Infrastructure** MobiCom 2026
  *Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, **Yufeng Xia**, Mun Choon Chan, Myungjin Lee, Mahesh Marina*
  - **AI-RAN Infrastructure Sharing**: Developed Weaver, a system enabling opportunistic foundation model training on GPU-accelerated RAN infrastructure, achieving 2.1–4.2× higher training throughput while harvesting up to 83% of spare compute without impacting RAN performance.
- **EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System** MobiCom 2026
  *Ujjwal Pawar, Andrew E. Ferguson, **Yufeng Xia**, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic*
  - **Scalable RAN Emulation**: Developed EmuRAN, a cloud-based RAN emulation system capable of emulating 10,000 mobile devices and 10,000 base stations on 130 cloud instances, two orders of magnitude larger than existing solutions.
- **Enhanced Pneumonia Detection in Chest X-Rays Based on Integrated Denoising Autoencoders and Convolutional Neural Networks** 2024
  YUFENG XIA
  - **Research Focus**: Conduct research on integrating machine learning with medical diagnostics to improve cancer classification accuracy using denoising autoencoders.
- **CoSense: Auto-Quantization Based on Sensor Specifications for Compiler Optimizations** Under Review, 2025
  YUFENG XIA, PEI MU, ANTONIO BARBALACE
  - **Framework Development**: Developed a novel LLVM-based auto-quantization framework leveraging sensor specifications to enhance performance and reduce code size for embedded systems.
  - **Quantization Techniques**: Implemented quantization techniques tailored to sensor data, achieving significant improvements in execution speed and binary optimization.
  - **Evaluation**: Evaluated CoSense on ARM Cortex-M0 and Cortex-A53 platforms, demonstrating measurable improvements in power, performance, and area (PPA).

Awards

- **Edinburgh Award**: Recognition of professional growth and invaluable experiences gained as a Research Assistant.
- **School Club Activities**: Table Tennis Enthusiast and Team Member. Active participant in school club activities, contributing to team dynamics and skill development.
- **BUCS Competitor**: Competed at a national level in the British Universities and Colleges Sport competitions.
- **Table Tennis Achievements**: Winner of Edinburgh Championship group stage, 3rd place in competition, MVP award, and "Most Improved Player in the league".
- **Table Tennis Achievements**: Secured 1st place in 2024 Edinburgh Championship Table Tennis Competition.